

DCET Syllabus & Marks distribution with effect from the Year 2023
(As per C_20 Diploma Curriculum)

Govt. of Karnataka
Department of Collegiate and Technical Education

DCET SYLLABUS with effect from the Year 2023 As per C_20 Diploma Curriculum (Common to all Engineering Diploma Programmes)		
DURATION: 3 Hours		MAXIMUM MARKS:100
Sl.No	TOPICS	MARKS
1	ENGINEERING MATHEMATICS	20
2	STATISTICS & ANALYTICS	20
3	IT SKILLS	20
4	FUNDAMENTALS OF ELECTRICAL & ELECTRONICS ENGINEERING	20
5	PROJECT MANAGEMENT SKILLS	20
TOTAL		100

DETAILED SYLLABUS

1. ENGINEERING MATHEMATICS

- 20 Marks

Topics	Sub Topics	Marks
I. Matrices And Determinants	- Matrix and types - Algebra of Matrices (addition, subtraction, scalar multiplication and multiplication) - Evaluation of determinants of a square matrix of order 2 and 3. Singular matrices - Cramer's rule for solving system of linear equations involving 2 and 3 variables - Adjoint and Inverse of the non- singular matrices of order 2 and 3 - Characteristic equation and Eigen values of a square matrix of order 2	4
II. Straight Lines	- Slope of a straight line - Intercepts of a straight line - Intercept form of a straight line - Slope-intercept form of a straight line - Slope-point form of a straight line - Two-point form of a straight line - General form of a straight line - Angle between two lines and conditions for lines to be parallel and perpendicular - Equation of a straight line parallel to the given line - Equation of a straight line perpendicular to the given line	4

**DCET Syllabus & Marks distribution with effect from the Year 2023
(As per C_20 Diploma Curriculum)**

III. Trigonometry	• Concept of angles, their measurement, Radian measure and related conversions. • Signs of trigonometric ratios in different quadrants (ASTC rule) • Trigonometric ratios of allied angles (definition and the table of trigonometric ratios of standard allied angles say $900\pm\theta$, $1800\pm\theta$, $2700\pm\theta$ and $3600\pm\theta$) • Trigonometric ratios of compound angles (without proof) • Trigonometric ratios of multiple angles • Transformation formulae	4
IV. Differential Calculus And Applications	• Derivatives of continuous functions in an interval (List of formulae) • Rules of differentiation • Successive differentiation (up to second order) • Applications of differentiation	4
V. Integral Calculus And Applications	• List of standard integrals and Basic rules of integration • Evaluation of integrals of simple function and their combination • Methods of integration • Concept of definite integrals • Applications of definite integrals	4
		20

2. STATISTICS AND ANALYTICS

- 20 Marks

Topics	Sub Topics	Marks
I. Statistical Data Collection And Types	• Definition of data and classification (qualitative, quantitative discrete and continuous data) • Data collection tools i) Questionnaires. ii) Survey. iii) Interviews. iv) Focus group discussion • Data cleaning	3

**DCET Syllabus & Marks distribution with effect from the Year 2023
(As per C_20 Diploma Curriculum)**

II. Summarization of Data	• Descriptive statisticsviii) i) Data tabulation (Frequency table) ii) Relative frequency table. • Grouped data i) Bar graph ii) Pie chart iii) Line graph iv) Frequency polygon v) Frequency curve vi) Relative frequency polygon vii) Histograms viii) Box plot • Leaf-stem plot	6
III. Measure of Location And Dispersion	• Determination of central tendencies Range, Mean, Mode and Median for the data • Determination of absolute measures of dispersion for data like range quartile deviation, mean deviation, standard deviation and variance. • Skewness and kurtosis graphs	5
IV. Introduction To Python Programming	• Introduction to PYTHON. • Syntax of PYTHON. • Comments of PYTHON. • Data types of PYTHON. • Variables of PYTHON. • If-else in PYTHON. • Loops in PYTHON. • Arrays and functions in PYTHON.	6
		20

3 . IT SKILLS

- 20 Marks

Topics	Sub Topics	Marks
I. Introduction to Basics of Coding	1.1 Introduction to computer programming 1.2 Algorithms –With sufficient examples 1.3 Flowcharts – With sufficient examples 1.4 Execute simple programs 1.5 Introduction to Application development 1.6 Simple android application development	4

**DCET Syllabus & Marks distribution with effect from the Year 2023
(As per C_20 Diploma Curriculum)**

II. Design And Develop Web Pages	2.1 Basic web technologies ▪ Browser ▪ Web –Server ▪ Client-Server Model ▪ URL ▪ SEO techniques ▪ Domain names and domain name system. 2.2 Creating Web-pages with HTML5 – Static ▪ Introduction, Editors ▪ Tags, Attributes, Elements, Headings ▪ Links, Images, List, Tables, Forms ▪ Formatting, Layout, Iframes. 2.3 Formatting web pages with style sheets(CSS3). ▪ Introduction to CSS ▪ Inline CSS, Internal CSS, Classes and IDs ▪ div, Color, Floating, Positioning ▪ Margins, Padding, Borders ▪ Fonts, Aligning Text, Styling Links 2.4 Creating a web page dynamic using JavaScript. ▪ Dynamic web page and Introduction to JS ▪ Basic syntax ▪ Functions ▪ Events 2.5 Creating dashboards in websites.	4
III. Business Process Automation / ERP	3.1 Introduction to business process automation. 3.2 Organization structure and functions composition- Properties and applications ▪ Structure ▪ Types ▪ Functional Units 3.3 Workflows ▪ Introduction ▪ Components ▪ Use and use cases 3.4 Enterprise resource planning ▪ History ▪ Evolution ▪ Uses of ERP	4

Handwritten signature

**DCET Syllabus & Marks distribution with effect from the Year 2023
(As per C_20 Diploma Curriculum)**

IV. Introduction To Cloud And IoT Concepts	4.1 Fundamentals of cloud 4.2 Cloud service models ▪ IaaS (Infrastructure-as-a-Service) ▪ PaaS (Platform-as-a-Service) ▪ SaaS (Software-as-a-Service) 4.3 Cloud deployment types ▪ Public, ▪ Private, ▪ Hybrid ▪ Community Cloud 4.4 Cloud services: ▪ Google Drive - file storage and synchronization service developed by Google; ▪ Google docs- bring your documents to life with smart editing and styling tools to help you easily format text and paragraphs; ▪ Google Co-lab (Usage of Jupyter Notebook): <i>Colab</i> notebooks allow you to combine executable code and rich text in a single document, along with images, HTML, LaTeX, and more. ▪ Google App Engine: Google App Engine is a Platform as a Service and cloud computing platform for developing and hosting web applications in Google-managed data centers. Applications are sandboxed and run across multiple servers. 4.5 Working of IoT and IoT components (Only brief introduction and demonstration through videos) 4.6 Explain concept of Internet of Things with examples ▪ Smart home ▪ Smart city ▪ Smart farming	4
---	--	-------------------------



**DCET Syllabus & Marks distribution with effect from the Year 2023
(As per C_20 Diploma Curriculum)**

V. Cyber security And Safety	5.1 Introduction to Cyber security and cybersafety. ▪ Brief awareness on cyber safety measures ▪ Identification of basic security issues in mobile phones and personal computers ▪ Installation of Antivirus software ▪ Firewall concepts ▪ Browser settings ▪ Importance of privacy and Password policy (Best practices). 5.2 Common threats - Demonstration ▪ Phishing ▪ DoS attack ▪ Man in the middle attack ▪ Eavesdropping ▪ Spamming	4
	20	

4 . FUNDAMENTALS OF ELECTRICAL & ELECTRONICS ENGINEERING

- 20 Marks

Topics	Sub Topics	Marks
I. Electrical Safety	1. Electrical Symbols 2. Electrical safety • Identify Various types of safety signs and what they mean • Demonstrate and practice use of PPE • Demonstrate how to free a person from electrocution • Administer appropriate first aid to victims, bandaging, heart attack, CPR, etc. • Fire safety, causes and precautionary activities. • Use of appropriate fire extinguisher on different types of fires. • Demonstrate rescue techniques applied during fire hazard, correct method to move injured people during emergency • Inform relevant authority about any abnormal situation 1.1 Earthing: Types	2

Handwritten signature

**DCET Syllabus & Marks distribution with effect from the Year 2023
(As per C_20 Diploma Curriculum)**

II. Electrical Fundamentals	1. Describe the sources of electrical energy. 2. Electrical current, voltage, emf, potential difference, resistance with their SI units. 3. Mention the meters used to measure different electrical quantities. Identification Measuring devices • Ammeter • Voltmeter • Wattmeter • Ohmmeter • Digital Multimeter • Megger • Tong tester 4. Explain supply systems like AC, DC. • Relationship between V, I and R. (Ohms law) • Behavior of V, I in Series and Parallel DC circuits. • Describe open circuit, close circuit and short circuit • Equation to find the effective Resistances connected in series • Equation to find effective Resistances connected in parallel • Resistances connected series and parallel combinations • AC sine wave: Sinusoidal voltage, current, amplitude, time-period, cycle, frequency, phase, phase difference, and their units. • Electrical work, power and power factor, SI units, mention the meters used to measure them. 5. Electrical energy • SI units • Mention the meters used to measure them • Single phase and Three phase supply.	5
--	--	-------------------------



**DCET Syllabus & Marks distribution with effect from the Year 2023
(As per C_20 Diploma Curriculum)**

III. Protective Devices And Wiring Circuits	1. Protective Devices • Necessity of Protective Devices • Various Protective devices and their functions • fuse wire, • Glass cartridge fuse • HRC fuse • Kit-kat fuse • MCB • MCCB • RCCB • ELCB • Relay • Different types of electrician tools and their function. • Describe various wiring tools. • State procedure of care and maintenance of wiring tools. 1. Describe different types of wiring systems. • Surface conduit • concealed conduit • PVC casing capping 2. Wiring systems and their applications. 3. Describe the types of wires, cables used for different current and voltage ratings.	3
IV. Electrical Machines and Batteries and UPS	1. Transformer • working principle • Transformation ratio • Types and applications with their ratings 2. Induction motor • Single phase and three phase Induction motor. • Necessity of starters. • Describe DOL AND STAR-DELTA starters. 3. What are different causes and remedies for a failure of starter and induction motor. 4. Battery • Types of batteries (Lead acid battery, lithium, sealed maintenance free (SMF) battery, Modular battery). • Selection criteria of batteries for different applications. • Ampere-Hour Capacity. • Efficiency 5. UPS • List the types and applications • Selection criteria of UPS • Sizing of UPS	5

A.Y

**DCET Syllabus & Marks distribution with effect from the Year 2023
(As per C_20 Diploma Curriculum)**

V. Introduction to Electronic Devices and Digital Electronics	1. Compare Conductors, insulators and semiconductors with examples. 2. Identification of types and values of resistors-color codes. 3. PN junction diode • Symbol • Characteristics • Diode as switch. • Types of diodes and ratings • Applications 4. Rectifier • Need for AC to DC conversion • Bridge rectifier with and without C filter, • Rectifier IC. 5. Transistor (BJT) • Symbol • Structure • Working principle • Comparison of analog and digital signal • Digital systems, examples. • Binary numbers, Boolean identities and laws. • Digital system building blocks: Basic logic gates, symbols and truth tables. • IC-Definition and advantages. 6. Sensors • Concept • Types: Temperature, Pressure, Water, Light, Sound, Smoke, proximity Sensors, Flow, humidity, voltage, vibration, IR(Principle/working, ratings/ specifications, cost, and applications) 7. Actuators • Concept • Types and applications. • Relay as an actuator 8. Microcontroller • As a programmable device and list of real-world applications. • PLC and Their applications.	5
		20

[Handwritten Signature]

**DCET Syllabus & Marks distribution with effect from the Year 2023
(As per C_20 Diploma Curriculum)**

5. PROJECT MANAGEMENT SKILLS

-20 Marks

Topics	Sub Topics	Marks
I. Introduction	• Meaning of Project • Features of a Project • Types of Projects • Benefits of Project Management • Obstacles in Project Management • Project Management as Profession • Project Manager and His Role • Project Consultants • What is Operation? • Difference between Project and Operation. • What is "Process" in Project Management and Process Groups? • What is Scope? Difference between Project Group Objectives and Project Scope.	4
II. Project Administration	• Essentials of Project Administration • Project Team • Project Design • Work Breakdown Structure (WBS) • Project Execution Plan (PEP) • Contracting Plan • Work Packing Plan • Organisation Plan • Systems and Procedure Plan • Project Procedure Manual • Project Diary • Project Execution System • Project Direction • Communication in a Project • Project Co-ordination • Pre-requisites for Successful Project Implementation	4

A.V

**DCET Syllabus & Marks distribution with effect from the Year 2023
(As per C_20 Diploma Curriculum)**

III. Project Life cycle	• Phases of Project Life Cycle • Project Management Life Cycle (General) • Project Planning • Project Execution • Project Closure • Project Risks • Types of Risks: Illustrations • Risk Assessment Techniques with Illustrations • Project Cost Risk Analysis • Estimating Time and Cost Overrun Risks • Organization/Procedural/Systemic Reasons for Project Cost Overruns • Time Overruns	4
IV. Project Planning, Scheduling and Monitoring	• Nature of Project Planning • Need for Project Planning • Functions of Project Planning • Steps in Project Planning • Project Planning Structure • Project Objectives and Policies • Tools of Project Planning • Project Scheduling • Time Monitoring Efforts • Bounding Schedules • Scheduling to Match Availability of Manpower • Scheduling to Match Release of Funds • Problems in Scheduling Real-life Projects • Introduction • Situation Analysis and Problem Definition • Setting Goals and Objectives • Generating Structures and Strategies • Implementation • What is Project Evaluation? • Why is Project Evaluation Important? • What are the Challenges in Monitoring and Evaluation?	4

Handwritten signature

**DCET Syllabus & Marks distribution with effect from the Year 2023
(As per C_20 Diploma Curriculum)**

V. Project Control, Review and Audit	• Projected Control Purposes • Problems of Project Control • Gantt Charts • Milestone Charts • Critical Path Method (CPM) • Construction of a Network • Network Technique in Project Scheduling • Crashing Project Duration through Network • Project Review • Initial Review • Post Audit • Performance Evaluation • Abandonment Analysis • Objectives of Project Audit • Functions of Project Auditor • Project Audit Programme • Difficulties in Establishing Audit Purpose and Scope • Digital Technology trends in Project management • Cloud Technology, IoT, AR and VR applications in Project management, Smart Cities	4
		20

